

The deployed model, slice by slice

What predicts each part of the target file, and how well

August 18, 2026

Why this deck exists

The target file is **not one forecasting problem**. It is three, and they need opposite things:

- a station with 18 months of its own history, extrapolated 1–6 months forward;
- a station that has *never* been observed, predicted for a period we already have labels for elsewhere in the network;
- a station never observed, *and* beyond the horizon.

Each slice gets its own estimator. This deck shows all three, with the measured number next to each design decision.

How everything here is measured

GEH < 5 — the traffic-engineering agreement statistic,

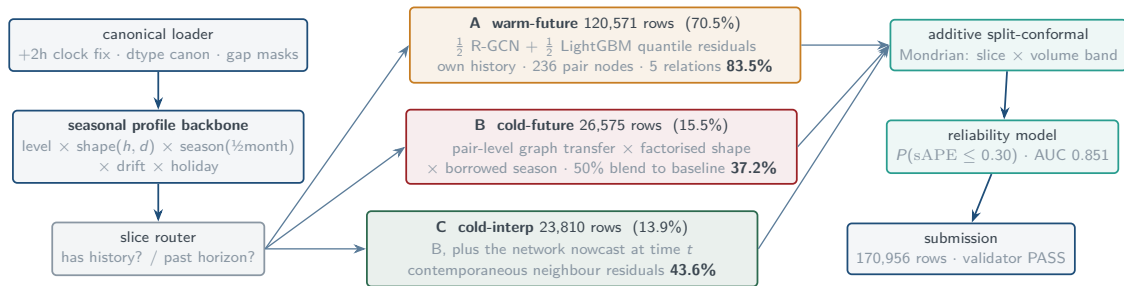
$$\text{GEH} = |M - C| \sqrt{2/(M + C)}$$
reported as the % of hours passing.

Two evaluation setups, **never comparable to each other**:

- **protocol** — 4-fold nested *station* CV. The only way to score cold start.
- **deployment arm** — fit 2024 + Jan–Apr 2025, score May–Jun 2025. Mirrors the real task for warm stations.

The map

One picture: what predicts what



Bold figures are GEH<5: warm on the deployment arm, cold on the 4-fold protocol — **never compare across the two**. Every slice shares one backbone; the slice-specific model only ever predicts a *residual* on top of it.

The three slices, side by side

	A warm-future	B cold-future	C cold-interp
rows	120,571 (70.5%)	26,575 (15.5%)	23,810 (13.9%)
period	Jul–Dec 2025	Jul–Dec 2025	Jan 2024 – Jun 2025
own history	18 months	none	none
network observed at t ?	yes	no (future)	yes
level from	own data	graph donors	graph donors
shape from	own data	matrix factorisation	matrix factorisation
season from	own data	corridor donors	corridor donors
shock at t from	—	—	network median
GEH<5	83.5% (deploy)	37.2% (protocol)	43.6% (protocol)
MAE	90.7	332.1	290.5
fold spread (sd)	0.6	5.2	5.1

The dominant lever is different on each side

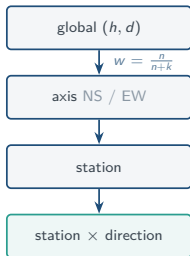
Oracle substitution — hand the model one perfect component. On warm-future a perfect **shape** is worth +9.3 GEH points, a perfect level only +1.7. On cold-interp it reverses: perfect **level** +18.5, perfect shape +2.0. Cold-start intuition applied to the warm slice sends you the wrong way.

The shared backbone

The backbone — structure first, learning second

Multiplicative decomposition, fitted in $\log(1+y)$ space

$$\log(1+y_{p,t}) = \underbrace{\ell_p}_{\text{level}} + \underbrace{\phi_{p,h,d}}_{\text{hour} \times \text{dow}} + \underbrace{\sigma_{p,b(t),h,d}}_{\text{seasonal shape}} + \underbrace{\delta_p}_{\text{year drift}} + \underbrace{\gamma_{\text{hol}}}_{\text{holiday}} + r_{p,t}$$



Each level shrunk toward its parent, $k_{\text{shape}}=3$, $k_{\text{season}}=1$. A thin cell degrades smoothly instead of switching hard.

The smoothed form of FHWA month-of-year adjustment factors.

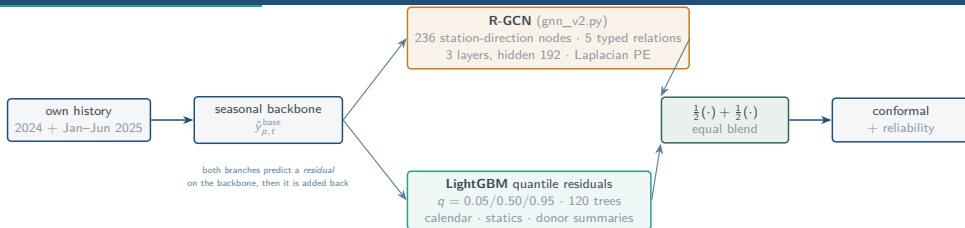
The seasonal kernel is the single biggest modelling gain

The profile is not noisy — it is *dated*: the same months a year apart disagree by 0.157 log against a 0.097 within-year noise floor. A target week's profile comes from a **half-month seasonal kernel** (24 bins), not pooled history:

profile estimator	error
2024 only	0.1412
month kernel $w=2$ (deployed)	0.1423
pooled (previous deploy)	0.1702
recency, 365d half-life	0.1741
2025 only (most recent)	0.2434

Slice A — warm-future

Slice A — two models on one backbone, averaged



Deployment arm, May–Jun 2025

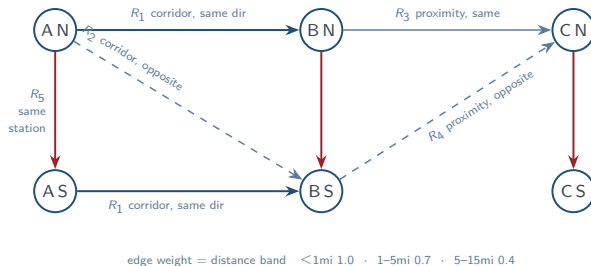
configuration	GEH<5	MAE
pooled backbone + LGB(350)	79.4%	102.7
seasonal backbone	82.7%	93.4
+ LightGBM (120 trees)	82.9%	92.0
R-GCN (hidden 192, +base)	83.4%	90.9
mean(R-GCN, LightGBM)	83.5%	90.7
oracle pair × h × d × season	88.6%	67.7

Why exactly 50/50, and not tuned

w _{GNN}	May–Jun	Mar–Apr	mean
0.00	82.92	83.41	83.17
0.25	83.22	83.41	83.32
0.50	83.46	83.32	83.39
0.75	83.51	83.13	83.32
1.00	83.44	82.88	83.16

The two windows' optima point in *opposite* directions (0.75 vs 0.00). The equal blend is the only window-independent choice.

Inside the R-GCN — why the graph is built on *directions*



3 layers $\Rightarrow$ a 3-hop receptive field: A N sees C's state through B without anyone writing a 2-hop rule.

The measurement that forced this design

Held-out station shape error (log space, hour $\times$ dow):

axis-level fallback	0.3445
direction-code level	0.3360
graph neighbours, direction-matched	0.2296

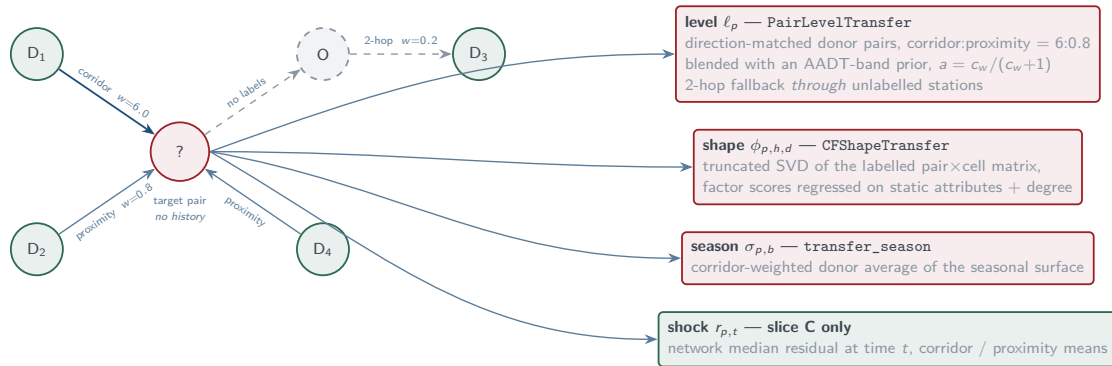
Along a corridor, *North at A is the same physical flow as North at B*. A station-level graph averages the two directions into one node and destroys exactly that.

Capacity is saturated

Twelve configurations, width 192–768: all land between 83.04% and 83.51%. The earlier "width is worth 10 points" was measuring a *missing input* — the backbone — not architecture.

**Slices B and C — stations never
observed**

Cold start — three components, three different transfers



D = donor with labels **O** = station present in `network_edges.csv` but never observed and never a target
— 9 of the 127 stations. The level transfer routes *through* them; nothing else does.

What each cold component is worth

Protocol, 4 folds — component ladder

cold-interp	GEH<5	MAE
backbone alone	36.2%	332.4
+ pair-level transfer	43.2%	291.3
+ factorised shape	43.6%	290.5

cold-future

backbone alone	33.5%	350.2
+ pair-level transfer	37.0%	333.2
+ factorised shape	37.2%	332.1

The factorised shape barely moves the mean but cuts fold spread (sd 6.88 $\rightarrow$ 5.11). On cold slices, *variance* is the real problem.

Level is where the cold gain came from

Held-out pair level error, mean $|\log|$:

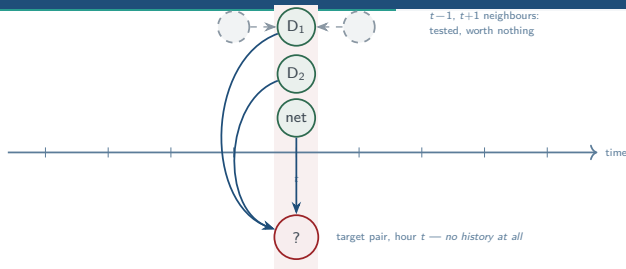
old path (donor blend + LightGBM)	0.3942
raw donor blend, no LightGBM	0.3859
direction-matched pair donors	0.3378

Two things carry it, and *neither is distance*: direction-matched *pair* donors instead of a direction-averaged station level, and a corridor:proximity ratio of 7.5:1 rather than 1.5:1.

Guard rail

On cold slices the residual learner is fitted on warm rows and does not transfer (MAE 285.7 vs 277.3 for the donor baseline alone). Its output is blended **50% back toward the backbone** rather than trusted.

Slice C only — the network nowcast



Slice C sits *inside* the labelled window, so the network is observed at the predicted hour. A regional shock — weather, an incident, an event — hits every station's residual at once: corridor correlation 0.84, proximity 0.62.

Do neighbour lags help? Measured: no

neighbour features	R^2	win rate
t only (deployed)	0.4882	—
$t-1, t$	0.4898	49%
$t-2, t-1, t$	0.4906	54%
$t-1, t, t+1$	0.4903	49%
$t-2 \dots t+2$	0.4912	52%

Ridge, 69 stations, temporal hold-out. Coin-flip win rates on a 0.49 base — hour-to-hour persistence already sits inside the profile.

Intervals, reliability, robustness

Intervals and the per-row reliability score

Additive split-conformal, Mondrian by slice $\times$ volume band

Offsets fitted on strictly out-of-time, out-of-station folds:

slice	coverage	width	Winkler
A	0.900	420	664
B	0.900	1156	1505
C	0.900	1353	1668

Raw quantile heads are badly overconfident exactly where the model knows least.

Multiplicative scaling exploded: a $\sim 20\times$ factor gave widths of 16,000 on a 622-vehicle forecast. Additive offsets are bounded by observed residuals.

Reliability: $P(\text{sAPE} \leq 0.30)$, isotonic-calibrated

features	AUC	Brier
slice + prediction + width	0.794	0.179
+ calendar + history depth	0.851	0.153

The original feature set could not express *when* the model is weak. Hour, day-of-week, month, holiday and night flags and per-pair history depth can.

A train/serve skew this exposed

History depth counted from the full panel gave held-out "cold" stations thousands of rows at fit time and zero at serve time — cold reliability collapsed to 0.06. Counts now come from each fold's own rows. The skewed version *scored better* (AUC 0.925); the implausible 0.06 is what exposed it.

Where the model is weak — and it is not where you would guess

Weakest slices (vs 72.3% overall, protocol)

slice	GEH<5	Δ
post-outage (1st hour after ≥ 24 h gap)	0.1%	−72.2
night hours	60.6%	−11.7
limited-history pair	64.9%	−7.4
holiday ± 1 day	66.3%	−6.0
winter	68.0%	−4.3

Strongest slices

slice	GEH<5	Δ
no corridor neighbour	76.0%	+3.7
summer	74.9%	+2.6
unreliable AADT band	74.5%	+2.2
peak hours	73.1%	+0.8
functional_class missing	72.2%	−0.1

Missing *context* does not hurt — missing *history* does

Missing functional_class, an unreliable AADT band and no corridor neighbour all score at or above the overall number.

The post-outage row is not a forecasting failure

On the first recorded hour after a ≥ 24 h gap the true value has median 67 veh/h where the model predicts ~ 613 — the sensor returns mid-hour and the record captures a *fraction* of the hour. No forecast can match a partial count. **A question for DeIDOT** (deldot_questions.pdf, Q8).

How much is left

Ceilings — what is actually attainable

slice	now	perfect profile	+ true network shock	
warm-future (deployment arm)	83.5%	88.6%	91.2%	85% target in reach
warm-future (protocol)	72.3%	88.4%	91.8%	protocol is pessimistic by design
cold-interp (protocol)	43.6%	80.5%	85.8%	oracle only
cold-future (protocol)	37.2%	not yet measured		open gap

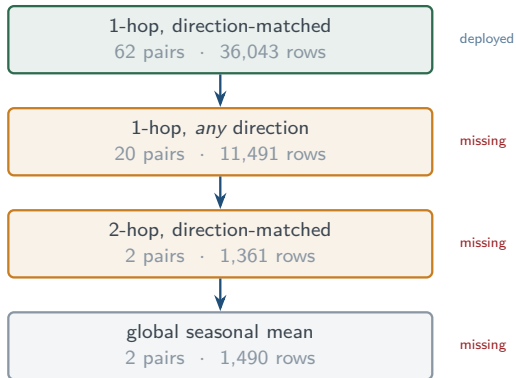
Reading the warm row

5.1 points of headroom to the oracle, and the 85% industry agreement target sits inside it. That headroom is *shape*, not level — which is why the seasonal kernel paid and level work did not.

Reading the cold row honestly

80.5% is what a *perfect deterministic profile* would score — not what is achievable. Cold level is bounded by what aadt_band plus the graph can say. Published work on this problem class reports 52.6% GEH<5 *using probe speeds, weather and coordinates that the supplied data does not contain*.

One gap found while preparing this deck



The fallback chain now in `transfer_season`. Before this change only the first rung existed and everything below it fell through to *zero seasonal correction*.

20 cold pairs — 12,247 rows, 7.2% of the submission — were getting no seasonal correction at all

Typical magnitude of the term: $0.110 \log (\approx 11\% \text{ of volume})$. After the fix, **zero rows** are left without one.

The fallback should ignore direction

Mean $|\log|$ difference between two seasonal surfaces:

corridor neighbour, <i>same</i> direction	0.0506
corridor neighbour, <i>opposite</i> direction	0.0654
same station, opposite direction	0.0715
random pair	0.2808
magnitude of the term itself	0.1097

An opposite-direction corridor donor (0.0654) lands closer to the truth than leaving the term at zero (0.1097) does.

...and what fixing it was actually worth

4-fold protocol, matched runs	GEH < 5		MAE	fold sd
cold-interp, deployed config	43.58 → 43.48	−0.10	290.5 → 289.6	5.11 → 5.27
cold-future, deployed config	37.22 → 37.32	+0.10	332.1 → 331.4	5.24 → 5.00
cold-interp, backbone only	36.23 → 36.62	+0.39	332.4 → 331.0	5.93 → 5.06
cold-future, backbone only	33.50 → 34.00	+0.50	350.2 → 349.1	5.38 → 4.78
warm-future, both arms	unchanged to 2 d.p. — the change cannot reach warm rows			

On the deployed cold config, the GEH effect is inside noise

Per fold on cold-interp:

+0.90/ − 1.20/ + 0.20/ − 0.30. MAE improves on 9 of the 10 cold arms and fold spread narrows on most, but nothing here moves the score.

Why it was still worth doing — and why it is small

A term silently set to zero for 7.2% of the submission is a structural defect, and it is now gone at no measured cost.

It is small for a reason we already knew: a *perfect* shape is worth only $\approx +2$ GEH on the cold slices. Partially recovering a shape-like term cannot be worth much there — the cold lever is **level**.

Is the network graph fully used?

consumer	1-hop	2-hop	routes through unlabelled stations
PairLevelTransfer (cold level)	yes, direction-matched	fallback only	yes
neighbour_features (shock)	yes	fallback only	yes
GraphShapeTransfer (hour×dow)	yes	no	—
transfer_season	yes	no	—
R-GCN (gnn_v2)	yes, 5 relations	yes, 3 layers	no — not in the node set

The unlabelled intermediate station

9 of the 127 stations in `network.csv` are neither observed nor targets, and **22 of the 186 edges** touch one. The level and shock paths hop *through* them correctly; the R-GCN builds its node set from observed $\cup$ target pairs, so those 22 edges silently vanish.

Cost today $\approx$ zero — but it would matter for a graph-based cold model.

2-hop for *shape*: measured, not worth it

The 22 held-out pairs across 4 folds that fall through to the direction-global shape:

estimator	error	wins
direction-global (deployed)	0.3365	—
2-hop direction-matched	0.3153	50%
50/50 mix	0.2740	68%

An oracle shape is worth $\approx +2$ GEH on cold-interp and this touches $\approx 18\%$ of cold pairs $\Rightarrow$ **about +0.1 GEH**. Not worth the code.

Ranked next steps

1. **Seasonal fallback chain** — **done**. 12,247 rows no longer carry a zero seasonal term; GEH-neutral, MAE marginally better, warm untouched.
2. **Oracle decomposition for cold-future** — 15.5% of the scored rows have no measured ceiling, so we cannot say whether further cold work is capped or open.
3. **Add the 9 unlabelled stations to the R-GCN node set** as feature-less relay nodes — only matters if a graph model takes over a cold slice.
4. *Not* 2-hop shape transfer — measured at $\approx +0.1$ GEH.

Scope statement

Everything in this deck is an **analytical estimate** produced from the supplied anonymised data. None of it is an engineering-grade operational, safety, capacity or design recommendation, and station identities were never recovered or inferred.

Questions